

Perfect Circle — Functional Design

Prepared by: Manya Bansal · Sep 27, 2026 · Version 0.1

The game has 10 screens around one shared play canvas; every shape, mode, duel and daily challenge reuses the same draw, validate, score and result loop.

Screens

Screen	What it shows	Main actions
Home	Today's daily challenge card, current streak, Play and Duel buttons	Start daily, play, duel
Setup	Shape picker (circle, square, triangle, star) and mode picker	Choose shape and mode, start
Play	Full-screen canvas, centre dot, shape hint, live score, timer in time limit mode	Draw
Result	Score, verdict line, stroke vs ideal overlay, new-badge pop-up	Retry, change mode, share
Daily	Challenge number, shape, mode, one attempt, then result and share	Play once, share
Duel lobby	Create room (code + QR) or join with code; both nicknames when joined	Create, join, ready
Duel play	Countdown, canvas, opponent's live progress bar, round score	Draw each round
Split-screen duel	Screen split in two halves, one dot each, shared countdown	Two players draw at once
Profile	Streak, best streak, badge grid, history graph with filters	Filter graph, sign in, edit nickname
Settings	Theme, sound, off-hand default, data export and reset	Change settings

Core play flow

1. The player picks a shape and mode on Setup, or opens the daily challenge.
2. Play shows the dot and a faint shape hint for 1 second.

3. The player presses and draws; the live score updates on every stroke point.
4. The attempt ends when the stroke completes one full turn around the dot, or when the player lifts their finger or mouse.
5. Validation runs (next table). A failed attempt shows the reason and a Retry button.
6. A valid attempt is scored, saved to history, checked against badges, and shown on Result.

Validation rules

Rule	Threshold	Message shown
Too close to the dot	Any point within 45 px of the dot	Too close to the dot. Try again.
Wrong direction	Stroke goes back more than 0.45 rad (about 26 degrees)	Wrong way. Keep going in one direction.
Shape not closed	Stroke covers less than 93% of a full turn	Close the shape. Try again.
Too slow	Attempt longer than 10 s (classic) or the mode's limit	Too slow. Try again.
Too short	Fewer than 10 stroke points	Draw a bigger shape.

Scoring by shape

Every stroke is first resampled to 128 evenly spaced points so fast and slow drawers are scored the same way.

- Circle: the ideal radius R is the mean distance of the points from the dot. The error d is the mean absolute difference between each point's distance and R .
- Square, triangle, star: the game builds the ideal shape centred on the dot, then finds the size and rotation that best fit the stroke (rotation searched in 1 degree steps). The error d is the mean distance from each stroke point to the nearest edge of that ideal shape, and R is the ideal shape's mean radius.
- Corner coverage (square, triangle, star only): each corner of the ideal shape must have a stroke point within 12% of R . The score is multiplied by the share of corners reached, so a circle drawn for a square scores low.

$$\text{score} = \max\left(0, 100 \times \left(1 - k \cdot \frac{d}{R}\right)\right) \times \text{corner coverage}$$

The strictness k is 2.5 for all shapes in v0 and will be tuned during business testing.

Game modes

Mode	Behaviour	How results are stored
Classic	No countdown; 10 s safety limit	mode = classic
Time limit	Countdown of 5, 3 or 2 s shown as a shrinking ring; attempt fails at 0	mode = timed, limit in seconds
Disappearing ink	Each stroke segment fades to invisible 0.4 s after it is drawn; the full stroke reappears on Result	mode = ink
Moving dot	Dot moves on a slow figure-eight path (about 30 px/s); each point is scored against the dot's position at that moment	mode = moving
Non-dominant hand	Player ticks "I'm using my other hand" before playing; can combine with any mode above	off_hand = true

Duels

Online duel on two devices:

1. Player A taps Create room and gets a 4-character code and QR code.
2. Player B taps Join and enters the code or scans the QR.
3. Both see each other's nickname and tap Ready.
4. The local server on the host PC picks the shape and mode for the round and starts a synced 3-2-1 countdown.
5. Both draw at the same time; each sees the other's progress bar filling as they draw.
6. Each device sends its resampled stroke; the server scores both and announces the round winner.
7. Rounds repeat until one player wins 2 of 3. A tie in a round goes to the faster finish.
8. Both see the final result with a Rematch button. If a player disconnects for more than 15 s, the other wins.

Split-screen duel on one device: the screen splits into two halves, each with its own dot. Each touch is assigned to the half where it started, so two fingers can draw at once. On a laptop with one mouse, players take turns on the same shape and the higher score wins.

Daily challenge

- The shape and mode come from the date: a hash of the UTC date string picks from the list, so every device gets the same challenge without asking the server.
- The challenge number is the count of days since launch day.
- One scored attempt per day; later tries are practice and are not counted.
- The share text copies to the clipboard or opens the phone's share sheet. Its five squares describe five equal parts of the stroke: green under 3% error, yellow under 7%, red above.

Perfect Circle #42 Star, disappearing ink
91.8% 
Streak: 12 days

Progression

- Streak: goes up by 1 when the daily challenge is completed on a UTC day, and resets to 0 if a day is missed. The best streak is kept.
- History graph: one dot per attempt, a line for each day's best score, and a 7-day moving average, filterable by shape, mode and off-hand.

Badge	How to earn it
First circle	Complete any valid attempt
Sharp eye	First score of 90% or more
Nearly perfect	First score of 95% or more
Human compass	First score of 99% or more
Shape master	85% or more on all four shapes
Ink ghost	90% or more in disappearing ink mode
Moving target	85% or more in moving dot mode
Other hand	80% or more with non-dominant hand on
Week streak	7-day daily streak
Month streak	30-day daily streak
First blood	Win a first duel
Duelist	Win 10 duels